APBR-SAM: Adaptive Perturbation Budget Reallocation for Post-Training N:M Structured Pruning

Abstract

N:M structured pruning removes weights through discrete support decisions, whereas Sharpness-Aware Minimization (SAM) trains with continuous local perturbations. Connecting these stages requires information that persists beyond an individual gradient direction and a selection rule that accounts for interactions between deleted weights. We propose APBR-SAM, which learns positive coordinate scales through perturbation-energy feedback while maintaining a fixed global L_2 radius. These scales retain a history of perturbation allocation and express subsequent weight deletions in normalized coordinates. To use this geometry with second-order reconstruction, we introduce a geometry-first conditional selection procedure. Within each 2:4 group, the smallest geometry-normalized weight determines the first deletion. A virtual OBS update then adjusts the remaining weights and inverse curvature, allowing the second deletion to reflect the effect of the first. Once both positions are selected, a single joint OBS update compensates them from the current physical weights. This design gives training geometry an explicit role in initiating support selection and uses conditional reconstruction to complete it, connecting historical perturbation allocation to one-shot structured pruning.

I. Introduction

Post-training N:M pruning converts a dense network into a structured sparse model by retaining N weights in every group of M. This pattern supports structured sparse execution [1], while imposing a local constraint on which connections survive. Dense optimization determines the weights available for pruning; support selection determines which of them are removed. We study how information learned during dense optimization can participate directly in this later decision.

Sharpness-Aware Minimization (SAM) provides a natural setting because it explicitly probes parameter changes during training [2]. Its perturbation follows the current mini-batch gradient within a prescribed local radius. An N:M deletion has a different structure: selected coordinates move to zero, their displacement magnitudes depend on the weights, and their locations obey a discrete support constraint. The mismatch concerns direction, scale, and support. The training probe explores both future retained and future removed coordinates, whereas the deletion acts on a selected support. A persistent description of coordinate exploration can carry information across these two stages.

APBR-SAM learns this description through adaptive perturbation budget reallocation. Positive coordinate scales reweight the first SAM gradient before the perturbation is normalized to a fixed global L_2 radius. Global energy feedback increases the relative scales of coordinates receiving below-average perturbation energy and decreases those receiving above-average energy. Applied recursively, this feedback produces a state that survives individual mini-batches. Dense training therefore returns both weights and matched coordinate scales, while retaining SAM's two-gradient optimization procedure.

Using this state for pruning requires connecting two measurements. APBR expresses the magnitude of deleting coordinate i as $|w_i| \exp(-s_i)$ in the learned geometry. OBS uses calibration activations to measure the quadratic reconstruction cost and the ability of other weights to compensate [3]. The first measurement supplies training history; the second supplies activation-dependent interactions. Our design gives each a specific role: geometry initiates a support decision, and reconstruction determines how that decision is completed.

In each 2:4 group, the APBR score selects the first deletion. We virtually apply its OBS update to the remaining weights and effective inverse curvature. The second deletion minimizes the resulting conditional OBS cost, so its selection reflects the predicted effect of the first removal. The selected pair then receives one joint OBS update from the physical state at the group entrance. Conditional updates and joint reconstruction follow established OBS principles [3]; APBR provides the training-dependent first choice.

The resulting pipeline connects historical perturbation allocation to structured reconstruction. Geometry answers where deletion begins, conditional curvature identifies its least costly partner, and joint compensation produces the sparse weights. This separation also defines a direct experimental comparison: change the first selection signal while keeping the dense parent, calibration inputs, conditional update, and final compensation fixed.

Our contributions are:

- A persistent coordinate-geometry mechanism that reallocates perturbation energy under a fixed global L_2 radius and carries the resulting training state into post-training support selection.
- A geometry-first conditional pruning procedure: APBR selects the initial deletion, the next decision uses its predicted reconstruction effect, and one joint physical OBS update compensates the selected pair.
- A controlled one-shot 2:4 evaluation protocol that isolates the initial geometric decision through matched dense parents, calibration inputs, and reconstruction settings.

II. Related Work

Perturbation geometry during dense training. SAM uses a two-gradient update to optimize loss in a local parameter neighborhood [2]. ASAM accounts for parameter rescaling [4], while SSAM restricts perturbations through Fisher-guided or dynamically updated masks [5]. GEAR-SAM uses historical gradient energy to allocate a fixed global perturbation budget across blocks [6]. APBR-SAM maintains positive coordinate scales through perturbation-energy feedback and saves this state with the dense model. The resulting geometry guides a structured deletion decision together with activation-based reconstruction.

Preparing dense models for compression. CrAM incorporates a compression operator into optimization to improve tolerance to subsequent compression [7]. AdaSAP applies importance-dependent perturbations before channel removal and sharpness-aware optimization afterward [8]. These approaches connect robust optimization with compression. APBR-SAM retains dense two-pass optimization and performs support selection after training. The information transferred between stages is a persistent coordinate geometry that expresses deletion relative to the learned scales.

Post-training selection and second-order compensation. Magnitude pruning ranks weight sizes [9], and Wanda combines magnitude with input-activation norms [10]. SparseGPT couples support selection with sequential second-order compensation [11]; ROSE modifies its processing order using estimated column and block pruning losses [12]. OBC/ExactOBS develops greedy OBS selection with conditional inverse-curvature updates and joint compensation for groups of weights [3]. Our procedure uses these reconstruction tools with two distinct selection signals. Geometry-normalized magnitude chooses the first deletion; virtual OBS conditioning supplies the state for choosing its partner. The pair receives one joint update over all currently active coordinates.

III. Method

APBR-SAM transfers a persistent training state into a two-stage support decision. The learned geometry selects the first deletion in each 2:4 group. A virtual reconstruction step then determines the state in which the second deletion is evaluated. Once both positions are selected, joint OBS commits their compensation to the physical weights.

A. Learning persistent perturbation geometry

The training state should describe how coordinates have been explored across mini-batches. Let $w_t \in \mathbb{R}^d$ collect the trainable parameters and let $\mathcal{L}_{B_t}$ denote the loss on mini-batch B_t. Standard SAM computes

$$g_{1,t} = \nabla_w \mathcal{L}_{B_t}(w_t), \qquad \epsilon_t^{\text{SAM}} = \rho \frac{g_{1,t}}{\|g_{1,t}\|_2}. \tag{1}$$

APBR adds positive coordinate scales through the log-state

$$a_{i,t} = \exp(s_{i,t}), \qquad A_t = \text{diag}(a_{1,t}, \ldots, a_{d,t}), \tag{2}$$

initialized with $s_0 = 0$. The state has a fixed correspondence to parameter names, shapes, and coordinates. It covers all trainable parameters, including BN affine parameters and biases; pruning subsequently acts on the eligible weight tensors.

To change the allocation of the perturbation budget while preserving its global magnitude, we use

$$\epsilon_t = \rho \frac{A_t^2 g_{1,t}}{\|A_t^2 g_{1,t}\|_2}. \tag{3}$$

The numerator sets the relative coordinate weights and the denominator fixes the realized L_2 norm. At initialization, this is the standard SAM direction. The second gradient is $g_{2,t} = \nabla_w \mathcal{L}_{B_t}(w_t + \epsilon_t)$, with the perturbation treated as fixed during differentiation. Both passes use the same augmented mini-batch. The model is restored to w_t before the base optimizer applies $g_{2,t}$.

The feedback signal is each coordinate's share of the perturbation energy:

$$p_{i,t} = \frac{\epsilon_{i,t}^2}{\sum_j \epsilon_{j,t}^2}, \qquad s_{i,t+1} = s_{i,t} + \gamma(1 - dp_{i,t}). \tag{4}$$

A coordinate receiving less than the reference share $1/d$ gains relative scale; a coordinate carrying more loses relative scale. This feedback redistributes future directional capacity and accumulates a history in s_t. The state therefore accompanies the final dense weights as information about the training trajectory. A candidate state is committed once after a successful parameter update; an all-zero perturbation leaves it unchanged.

B. Transferring geometry to a deletion score

Deleting coordinate i produces displacement $-w_i$. Expressing this displacement in the learned coordinate scale gives

$$I_i = \frac{|w_i|}{a_i} = |w_i| \exp(-s_i). \tag{5}$$

The score compares the size of a weight with its learned exploration scale. For a mask m and deletion displacement $\delta_m = (m - 1) \odot w$, the corresponding squared cost is

$$C_A(m) = \left\| A^{-1}\delta_m \right\|_2^2 = \sum_{i:m_i=0} I_i^2. \tag{6}$$

This gives the training state a concrete role in support selection: smaller scores identify smaller deletions in normalized parameter coordinates.

A second criterion enters when the surviving weights are allowed to reconstruct the layer output. Their ability to compensate depends on correlations between input activations. We use the geometry score to initiate the support decision, then use the conditional reconstruction cost to complete it. These two stages connect the saved training geometry with the activation information available at pruning time.

C. The physical reconstruction state

For an eligible layer, flatten each output row into a weight vector and collect calibration inputs as columns of $X \in \mathbb{R}^{K \times P}$. Convolutional inputs follow the stored unfold order. Define

$$H = \frac{2}{P} X X^\top, \qquad H_\delta = H + \delta I, \qquad \delta > 0. \tag{7}$$

For a row displacement Δw, the quadratic $\frac{1}{2}\Delta w^\top H_\delta \Delta w$ combines the mean squared output change with damping. The activation matrix and damping are fixed during the layer's pruning.

Groups are processed in their stored order. At a group entrance, F contains every coordinate still allowed to change, including the current group G and all later groups. Earlier groups have been frozen. The effective inverse curvature is

$$R = \left((H_\delta)_{FF}\right)^{-1}, \qquad G \subseteq F. \tag{8}$$

Here w_F denotes the current physical weights after compensation from earlier groups. With U the upper Cholesky factor of H_δ^{-1} and F an unprocessed suffix, the same matrix is $R = U_{FF}^\top U_{FF}$. The group block R_{GG} supplies local selection quantities, while $R_{:,D}$ later propagates compensation across the full active set.

D. Geometry-guided initial deletion

For the current quartet G, APBR selects the first deletion directly:

$$i^* = \arg\min_{i \in G} |w_i| \exp(-s_i). \tag{9}$$

The saved geometry is fixed throughout pruning. Thus the first choice connects the current physical weights to the coordinate scales learned during dense training.

The effect of this choice must be reflected in the second decision. We create a virtual state corresponding to deletion of i^*:

$$\bar{w}_j = w_j - \frac{R_{ji^*}}{R_{i^*i^*}} w_{i^*}, \qquad \bar{R}_{jk} = R_{jk} - \frac{R_{ji^*} R_{i^*k}}{R_{i^*i^*}}, \tag{10}$$

for $j, k \in F \setminus \{i^*\}$. The weight update predicts how the first deletion is reconstructed, and the Schur update describes the curvature after that coordinate is constrained. These quantities are held in a separate selection state.

E. Conditional second selection and joint compensation

The remaining three positions are compared in the virtual state. The second deletion is

$$j^* = \arg\min_{j \in G \setminus \{i^*\}} \frac{\bar{w}_j^2}{2\bar{R}_{jj}}. \tag{11}$$

Its score measures the additional quadratic cost of removing j after the first deletion. This is the point at which activation-dependent interactions enter the support decision. Both selection steps resolve exact score ties by the lower stored coordinate index.

Set $D = \{i^*, j^*\}$ and discard the virtual state. Joint OBS uses the weights and effective inverse curvature saved at the group entrance:

$$\Delta w_F = -R_{:,D} (R_{DD})^{-1} w_D, \qquad C_{\text{OBS}}(D) = \frac{1}{2} w_D^\top (R_{DD})^{-1} w_D. \tag{12}$$

This is the standard fixed-support quadratic reconstruction update [3]. It sets the two deleted coordinates to zero and adjusts the remaining active coordinates. The physical update is committed once, after which the entire quartet is frozen. The next quartet starts from weights that already include this compensation.

The three states have distinct roles: s carries training history, $(\bar{w}, \bar{R})$ predicts the effect of the first deletion, and (w_F, R) defines the physical compensation. Separating them prevents a virtual deletion from being counted again in the joint update.

F. End-to-end procedure

Algorithm 1 summarizes the complete path. Dense training saves unperturbed weights together with their matching geometry. Calibration then supplies a shared reconstruction metric for each layer. Each quartet receives a geometry-guided first choice, a conditional second choice, and one committed reconstruction. The resulting mask retains exactly two coordinates per complete quartet.

Algorithm 1 APBR-guided conditional OBS: dense training to 2:4 pruning

Require: training data, calibration inputs, weights w, radius ρ, feedback rate γ

$s \leftarrow 0$
for each dense-training mini-batch B_t **do**
  $A_t \leftarrow \text{diag}(\exp(s_t))$; $g_{1,t} \leftarrow \nabla_w \mathcal{L}_{B_t}(w_t)$
  if $\|A_t^2 g_{1,t}\|_2 > 0$ **then**
    Compute ϵ_t and candidate $\widehat{s}_{t+1}$ by Eqs. (3)–(4)
  else
    $\epsilon_t \leftarrow 0$; $\widehat{s}_{t+1} \leftarrow s_t$
  end if
  $g_{2,t} \leftarrow \nabla_w \mathcal{L}_{B_t}(w_t + \epsilon_t)$
  Restore w_t; apply the base optimizer using $g_{2,t}$; commit $s_{t+1} \leftarrow \widehat{s}_{t+1}$
end for
Save the matched dense weights and geometry; freeze geometry
for each eligible layer **do**
  Form H_δ from calibration activations; initialize a bool keep-mask to true
  for each complete quartet G in stored column order **do**
    $F \leftarrow$ all unfrozen columns; $R \leftarrow ((H_\delta)_{FF})^{-1}$
    for each output row **do**
      Save current physical w_F; $i^* \leftarrow \arg\min_{i \in G} |w_i| \exp(-s_i)$
      Form virtual $(\bar{w}, \bar{R})$ on $F \setminus \{i^*\}$ by Eq. (10)
      $j^* \leftarrow \arg\min_{j \in G \setminus \{i^*\}} \bar{w}_j^2 / (2\bar{R}_{jj})$
      $D \leftarrow \{i^*, j^*\}$; discard virtual state
      Compute Δw_F from saved physical w_F, R by Eq. (12)
      Commit $w_F \leftarrow w_F + \Delta w_F$; set $w_D \leftarrow 0$ and keep-mask$_D \leftarrow$ false
    end for
    Freeze all columns in G
  end for
  Retain any incomplete tail; validate the 2:4 mask and write back the layer
end for
return compensated sparse weights and keep-masks

IV. Theoretical Analysis

A. Training geometry properties

Equation (3) gives $\|\epsilon_t\|_2 = \rho$ whenever its numerator is nonzero. The energy shares also conserve the sum of log-scales:

$$\sum_i s_{i,t+1} = \sum_i s_{i,t} + \gamma \left(d - d \sum_i p_{i,t} \right) = 0. \tag{13}$$

Thus $\det A_t = \prod_i a_{i,t} = 1$ in exact arithmetic. This identifies the relative coordinate geometry while the global normalization fixes the realized perturbation budget.

For a fixed gradient with nonzero components, the energy shares are

$$p_i = \frac{\exp(4s_i + 2\log|g_i|)}{\sum_j \exp(4s_j + 2\log|g_j|)}. \tag{14}$$

Proposition 1 (Fixed-gradient feedback)**.** *For fixed $g_i \neq 0$, positive γ, and $\sum_i s_i = 0$, the feedback fixed point has $p_i = 1/d$ and*

$$s_i^* = -\tfrac{1}{2}\log|g_i| + \frac{1}{2d}\sum_j \log|g_j|, \qquad a_i^* \propto |g_i|^{-1/2}. \tag{15}$$

For $d > 1$, its linearized eigenvalue on the zero-sum subspace is $1 - 4\gamma$. The strict linear stability range is $0 < \gamma < 1/2$.

The fixed-gradient analysis explains how negative feedback balances the coordinate energy shares. Its derivation appears in Appendix A.

B. Direct geometry deletion

The geometry score has a separable interpretation. With weights and scales fixed, a direct N:M mask minimizes $C_A(m)$ by retaining the N largest I_i in each group. Exchanging a retained smaller score with a deleted larger score reduces the cost. For the initial decision in Algorithm 1, the same ordering selects the smallest normalized singleton displacement.

The associated loss bound follows the finite deletion segment. Let $\mathcal{L}$ be a fixed loss, with auxiliary model state fixed, and $g = \nabla\mathcal{L}(w)$.

Proposition 2 (Conditional bound for direct deletion)**.** *Fix w, a positive diagonal matrix A, and a family $\mathcal{M}$ of candidate masks. Suppose $\mathcal{L}$ is twice differentiable along every segment $w + t\delta_m$, and a single $\beta_A \geq 0$ satisfies*

$$\sup_{\substack{m\in\mathcal{M}\\ t\in[0,1]}} \left\| A\nabla^2\mathcal{L}(w + t\delta_m)A \right\|_2 \leq \beta_A, \tag{16}$$

where the matrix norm is spectral. Then every $m \in \mathcal{M}$ satisfies

$$\mathcal{L}(w + \delta_m) - \mathcal{L}(w) \leq \|Ag\|_2 \sqrt{C_A(m)} + \frac{\beta_A}{2} C_A(m). \tag{17}$$

Writing $\delta_m = Aq_m$ turns the quadratic Taylor term into a bound on $\|q_m\|_2^2$. Cauchy–Schwarz bounds the linear term by $\|Ag\|_2\|q_m\|_2$, giving Eq. (17).

Corollary 1 (Minimum common upper bound)**.** *For fixed w, A, g and the shared β_A, the function $U_A(c) = \|Ag\|_2\sqrt{c} + (\beta_A/2)c$ is nondecreasing for $c \geq 0$. If $\mathcal{M}$ consists of independent N:M groups with all other coordinates retained, keeping the N largest I_i in each group minimizes $U_A(C_A(m))$ over $\mathcal{M}$.*

Appendix B gives the complete deletion-segment argument.

C. Conditional selection and fixed-support reconstruction

The second stage has a direct interpretation in the physical quadratic objective. Fix the geometry-selected coordinate i. For a candidate partner j, block inversion gives

$$C_{\mathrm{OBS}}(\{i,j\}) = \frac{w_i^2}{2R_{ii}} + \frac{\left(w_j - \frac{R_{ji}}{R_{ii}}w_i\right)^2}{2\left(R_{jj} - \frac{R_{ji}^2}{R_{ii}}\right)} = \frac{w_i^2}{2R_{ii}} + \frac{\bar{w}_j^2}{2\bar{R}_{jj}}. \tag{18}$$

The first term is fixed once APBR selects i. Minimizing the second term therefore chooses the least costly partner among the three remaining positions under the current reconstruction objective. Both the compensated weight and the change in available reconstruction directions enter this decision.

For the resulting set D, joint reconstruction solves

$$\min_{\Delta w_F} \frac{1}{2}\Delta w_F^\top (H_\delta)_{FF}\Delta w_F \quad \text{subject to} \quad (\Delta w_F)_D = -w_D. \tag{19}$$

The Lagrange stationarity equation is $\Delta w_F = -R_{:,D}\lambda$. Substitution into the deletion constraint gives $\lambda = (R_{DD})^{-1}w_D$, yielding Eq. (12). Positive definiteness makes this fixed-support solution unique.

For the same deletion set, successive exact OBS updates with Schur conditioning satisfy the same quadratic constraints and recover the same solution [3]. Algorithm 1 uses that conditional state to choose the partner, then commits the fixed-support solution in one update. This connects the selection calculation directly to the physical compensation.

V. Experiments

Evaluation setting. The evaluation uses CIFAR-100 and the 100-class CIFAR-ResNet18 [13], with the fixed final dense checkpoint as the pruning parent. Each method targets 2:4 sparsity in the same eligible internal convolutional layers. Dense training uses 200 epochs. The stem, classifier, BN parameters, and biases remain dense.

A. Post-training endpoints

Table 1 provides the existing reference accuracies and the evaluation slot for APBR-guided conditional OBS. The SAM reference applies a magnitude mask, and Wanda [10] uses activation-aware scores; both are mask-only endpoints. SparseGPT [11] includes its native selection and OBS compensation. The pending APBR-SAM evaluation will follow the geometry-first and conditional-second procedure in Algorithm 1, including joint compensation.

Table 1: CIFAR-100 post-pruning accuracy. Completed reference rows retain their existing three-seed mean ± sample SD. PENDING denotes the evaluation of the geometry-first method defined in this paper.

Method	Accuracy (%)
SAM	55.54 ± 1.36
Wanda	56.48 ± 3.83
SparseGPT	77.26 ± 0.42
APBR-SAM	PENDING

The reference results identify reconstruction as an important part of the post-training endpoint: SAM and Wanda report $55.54 \pm 1.36\%$ and $56.48 \pm 3.83\%$ after mask application, while native SparseGPT reports $77.26 \pm 0.42\%$ after selection and compensation. The APBR evaluation combines support selection with reconstruction and measures its accuracy at the same post-pruning stage, without recovery training.

B. Geometry-guided selection

The mechanism comparison uses the same APBR dense parent, calibration inputs, damping, processing order, and joint compensation. The physical conditional-OBS control chooses its first deletion by singleton OBS cost. APBR-guided conditional OBS instead chooses it by the saved geometry score. Both then select the second position in the virtual conditional state. This comparison directly tests the role of the initial geometric decision.

Table 2: Planned same-parent comparison of the initial deletion rule. Both rows use conditional OBS for the second deletion and the same joint compensation.

Initial deletion rule	Accuracy (%)
Physical singleton OBS	PENDING
APBR geometry	PENDING

The evaluation will record mask agreement, physical layer-output reconstruction error, and final classification accuracy. Mask agreement locates changes in the selected support; reconstruction error measures their layer-level effect; classification accuracy measures the resulting network endpoint. Before evaluation, the resulting masks and compensated checkpoints will be frozen for a single final test.

VI. Conclusion

APBR-SAM carries information from dense optimization into structured support selection. Global energy feedback learns persistent coordinate scales under a fixed perturbation budget. At pruning time, those scales determine the first deletion in each quartet; the second decision uses the reconstruction state induced by a virtual OBS update. Joint compensation then reconstructs the layer for the selected support. This division of roles gives the training geometry a direct entry point into pruning while using activation correlations to complete and compensate the decision. The resulting method connects historical perturbation allocation, conditional support selection, and second-order reconstruction in a single post-training 2:4 pipeline.

A. Feedback Derivation

Squared perturbation coordinates are proportional to $a_i^4 g_i^2$, giving Eq. (14). A fixed point with $\gamma > 0$ requires $p_i = 1/d$. Thus $4s_i + 2\log|g_i|$ is constant. Enforcing $\sum_i s_i = 0$ yields Proposition 1.

Differentiating the shares gives $D_s p = 4(\mathrm{diag}(p) - pp^\top)$, so the update Jacobian is

$$J = I - 4\gamma d(\mathrm{diag}(p) - pp^\top). \tag{20}$$

At $p = \mathbf{1}/d$, $J = I - 4\gamma(I - \mathbf{1}\mathbf{1}^\top/d)$. Every zero-sum direction has eigenvalue $1 - 4\gamma$, giving $|1 - 4\gamma| < 1$ precisely when $0 < \gamma < 1/2$. The common-scale direction has eigenvalue one and is excluded by the zero-sum constraint. If a fixed gradient coordinate is zero while another is nonzero, its share is zero and its log-scale increases by γ each step; the component-wise nonzero assumption is required for this interior fixed point.

For the training perturbation ϵ in Eq. (3), the linearized loss change is

$$g^\top \epsilon = \rho \frac{g^\top A^2 g}{\|A^2 g\|_2} \le \rho \|g\|_2 , \tag{21}$$

by Cauchy–Schwarz. Standard SAM maximizes the linearized loss on the sphere; APBR-SAM uses the learned state to reweight that direction.

B. Pruning-Loss Bound: Complete Proof

This appendix proves Proposition 2 and Corollary 1 for the deletion displacement δ_m. The training perturbation ϵ in Eq. (21) is a different vector.

Proof of Proposition 2. Fix any $m \in \mathcal{M}$ and set $q_m = A^{-1}\delta_m$ and $h_m(t) = \mathcal{L}(w + t\delta_m)$. Since A is positive diagonal, it is invertible and $\delta_m = Aq_m$. Along the candidate segment,

$$\begin{aligned} h_m'(0) &= g^\top \delta_m = (Ag)^\top q_m, \\ h_m''(t) &= q_m^\top A \nabla^2 \mathcal{L}(w + t\delta_m) A q_m \le \beta_A \|q_m\|_2^2. \end{aligned} \tag{22}$$

The last inequality follows from the spectral-norm bound in Eq. (16). Define $r_m(t) = h_m(t) - h_m(0) - t h_m'(0) - (\beta_A/2)t^2\|q_m\|_2^2$. Its second derivative is nonpositive. The mean value theorem therefore makes r_m' nonincreasing; since $r_m(0) = r_m'(0) = 0$, it follows that $r_m(1) \le 0$. Thus

$$\mathcal{L}(w + \delta_m) - \mathcal{L}(w) \le (Ag)^\top q_m + \frac{\beta_A}{2}\|q_m\|_2^2. \tag{23}$$

Applying $(Ag)^\top q_m \le \|Ag\|_2 \|q_m\|_2$ and $\|q_m\|_2^2 = C_A(m)$ gives Eq. (17). The case $C_A(m) = 0$ has $\delta_m = 0$ and both sides equal zero. The same β_A applies to every mask by assumption, so the conclusion holds throughout $\mathcal{M}$. □

Proof of Corollary 1. For any $0 \le u \le v$,

$$U_A(v) - U_A(u) = \|Ag\|_2(\sqrt{v} - \sqrt{u}) + \frac{\beta_A}{2}(v - u) \ge 0. \tag{24}$$

For independent N:M groups, Eq. (6) is a sum of squared scores over deleted coordinates. In each group, keeping its N largest scores minimizes that group's contribution. Summing these groupwise minima minimizes $C_A(m)$ over the admissible family; monotonicity then minimizes the same function $U_A(C_A(m))$. Equal scores can yield multiple minimizing masks, resolved by the prescribed tie rule. The comparison holds with w, A, g, β_A fixed, and orders upper bounds rather than realized losses. □

C. Evaluation Protocol

The protocol uses the official CIFAR-100 training and test splits with 50,000 and 10,000 images, respectively. The dense parent is the fixed final checkpoint after 200 epochs. All methods use the same 19 eligible internal Conv2d layers, output-row-contiguous OIHW groups, and dense incomplete tails. The input stem, classifier, BN affine parameters, and biases are excluded from pruning.

The geometry-first evaluation uses the matched dense weights and log-scales, fixed training calibration identities, and frozen reconstruction settings. Calibration and pruning retain the dense parent's BN buffers. After all masks and compensated checkpoints are frozen, each final artifact is evaluated once on the test split. No recovery training is included. The accuracy slots for this method are marked PENDING in Section V.

At each group entrance, the current physical weights include updates committed by preceding groups. The active inverse in Eq. (8) is recomputed mathematically for the remaining free coordinates; its shared Cholesky-tail representation avoids a separate inverse for each output row. All exact selection ties use the lower stored coordinate index. After the selected pair is compensated, the full quartet is frozen before advancing.

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